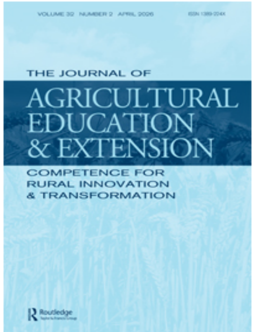

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Fostering innovation through farmer interactions: social networks and technology

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ABSTRACT

Purpose: This study finds that the impact of social networks, specifically interactions and knowledge sharing among farmers, influences the adoption of agricultural innovation.
Methodology: Using an extensive social network dataset from 1,392 households across 93 villages in Central and Northeast Thailand, the study investigates the adoption of five key rice production technologies. A multivariate probit regression model identifies factors influencing correlated adoption decisions, emphasizing various positions within farmers' networks.
Findings: Farmers occupying more central positions within the network, measured by various centrality metrics, are 4–9% more likely to adopt new technologies. This effect is especially pronounced among those who maintain a greater number of connections and play a key role in disseminating information.
Practical Implications: Farmers who actively share information with others are significantly more likely to adopt, underscoring that being a source of knowledge within the network has a more decisive influence than merely being a recipient. These findings highlight the importance of fostering active information sharing among farmers to accelerate innovation.
Theoretical Implication: The study presents the theoretically robust impacts of social networks on technology adoption decisions for multiple correlated technologies.
Originality: The study emphasizes the importance of outward-oriented social connections, showing their distinct influence on farmers' technology adoption behavior.

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

Introduction

There has been a concern that poverty and inequality in agricultural households persist despite rapid economic growth in developing countries. Thailand is an example of a country where agriculture has historically played a dominant role in industry but is currently experiencing a shortcoming in productivity. Recent statistics highlight that rural farms face multifaceted challenges, such as low average income and limited access to education, healthcare, and infrastructure, exacerbated by climate risks (Attavanich et al. 2019; World Bank 2022). The Thai government employed traditional price intervention with land reclamation, which successfully increased total productivity until the 1990s. However, as the potential of land ran out and the subsidies tightened government expenditure, continuous technological innovation emerged as a fundamental breakthrough. Technology innovation offers a hopeful and optimistic outlook for alleviating poverty in the agriculture sector, inspiring us to continue our investigation in this field (Faysse et al. 2020).

Specifically, Thailand's government launched a global standard for a Sustainable Rice Platform (SRP) and promoted SRP-themed technologies to induce climate-adaptive innovations in farm practices. Access to and adoption of this integrated production system not only affects input and output sustainability but is more affirmative of global demand, which is germane to farm income in Thailand (Takahashi, Muraoka, and Otsuka 2020). However, farmers are reluctant to make changes, especially when the costs and benefits are random, which commonly leads to lagged adoption of technologies in the developing world (Dercon and Christiaensen 2011; Koundouri, Nauges, and Tzouvelekas 2006). The problem becomes more complex when multiple interconnected options are involved, making it challenging for farmers to maximize their expected utility strategically.

Over the years, it has been discussed that socio-economic determinants, mainly the information exchange among individuals, facilitate farmers' adoption of technologies. The foremost empirical findings of Conley and Udry (2010) found how geographical and social neighbors influence the pathway of technology diffusion. Farmers do not make decisions unilaterally but collect information from multiple sources and update their knowledge, which can further change their attitude toward stochastic technologies. The government remains a leading actor in information dissemination; however, individuals also transmit their empirical findings after demonstrations in the field to convince or dissuade others. Thus, the number of information sources may reveal the level of social capital, which extends to resource accessibility, bargaining power, and innovation and adaptation strategies.

Since information is rarely symmetric in low – and middle-income countries, private information networks are expected to facilitate technology adoption (Hidayat, Noor, and Handono 2024). Existing literature on social networks and agricultural technology adoption has predominantly emphasized the economic returns derived from these networks. Some studies have also highlighted trust and risk mitigation as additional benefits of social networks for rural households (Hoang, Castella, and Novosad 2006). However, relatively few studies have explicitly investigated how the structural characteristics of social networks, particularly farmers' relative positions within networks based on knowledge exchange, may affect the adoption of agricultural innovations. To address this potential gap, this study examines how farmers' positions in knowledge-sharing networks might influence their decisions regarding technology adoption.

Specifically, this study aims to identify the determinants of sustainable technologies for rice production in Thailand, mainly when multiple correlated decisions are made simultaneously. Our primary goal is to examine the effect of social networks as a decisive factor in innovation, where a network is defined by individual farmers (nodes) and their interactions (edges). The more informative connections farmers have, the more likely they are to adopt these technologies, shifting from 'as usual' to a more sustained production practice (Maertens and Barrett 2013). In line with the approach of Foster and Rosenzweig (1995), we expect farmers will 'learn from others' to overcome information constraints in production. Specifically, we employ various network centralities to capture farmers' pre-existing position in a social group, which reflects structural dimensions of social learning.

In the study area, we identify five SRP-themed technologies that have the potential to enhance both short – and long-term productivity while minimizing environmental impact. Since multiple technological choices are considered simultaneously, it is necessary to account for the probabilistic correlations between them, as adopting one technology may influence the likelihood of adopting others. The empirical multivariate probit regression model enables the simultaneous comparison of multiple binary dependent variables. We specify the determinants of each technology, including farm characteristics such as land and labor, resource accessibility, and household characteristics that could significantly promote adoption. Furthermore, we incorporate four different social network centrality measurements to highlight the network effects in these decisions.

This study offers important methodological insights by demonstrating how diverse network indices can influence multiple correlated technologies. It provides a realistic view of how household characteristics and resource accessibility can encourage or limit adoption when introducing stochastic technologies. By incorporating the qualitative dimensions of information flow, the study reveals that active knowledge sharing in a social network can significantly promote adoption.

From a policy perspective, the findings suggest a crucial shift in focus from information receivers to information senders within technology adoption initiatives. It contributes to designing more effective strategies that resonate with the social dynamics of rural communities. This approach strengthens the reach and effectiveness of adoption efforts and offers valuable insights for extending similar initiatives to other developing countries.

Data and study area

This study focuses on central and northeastern Thailand, where rice production is the primary source of farm income. In 2022, a field study was conducted using a structured questionnaire asking about the social networks of individual households. In the nine provinces shown in Figure 1, we randomly selected 93 villages and employed snowball sampling to collect data from 1,392 agricultural households.

Snowball sampling is a widely used method in social network analysis when examining the entire network is inefficient and costly. The technique enables researchers to identify participants through existing connections, making it particularly effective for

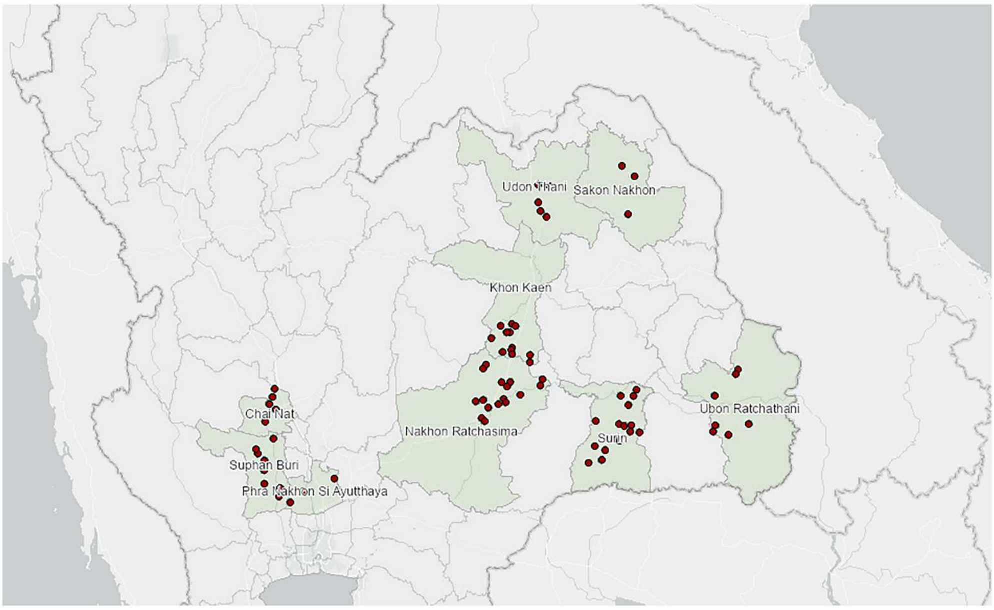

Figure 1. Study area of nine provinces in Thailand.

reaching hard-to-access populations (Biernacki and Waldorf 1981). In this study, we specifically asked respondents to name other households with whom they actively share agricultural knowledge and information. We initiated the sampling process with village leaders, presumed to be among the most central nodes, and then expanded the sample to 16 additional households directly connected to these initial nodes. It also has advantages over other sampling methods in that most nodes are relatively central to the network; however, it may underrepresent minority groups if drawn from heterophilic networks (Wagner et al. 2017). We alleviate this potential representative bias by using sufficient sample sizes and controlling for multiple household characteristics.

Agricultural production in the study area mainly focuses on rice and glutinous rice, where over 70% of the sample produced a single rice crop. Notably, farmers commonly rotate between rice and glutinous rice, a 'rotation of monoculture,' which means they plant in – and off-season rice. A household growing a monoculture suffers from low

Table 1. The number of sampled households by province.

Province	Central Region	The Number of Villages	The Number of Households
Ayutthaya	Yes	4	61
Chainat	Yes	6	92
Khon Kaen	No	14	191
Nakhon Ratchasima	No	17	262
Sakon Nakhon	No	3	45
Suphan Buri	Yes	12	158
Surin	No	22	349
Ubon Ratchanthani	No	9	142
Udon Thani	No	6	92
Total		93	1392

and significant variations in productivity and is more susceptible to risk, especially when exposed to a highly volatile climate shock (World Bank 2022).

As shown in Table 1, there is a structural disparity in farm size between regions; specifically, farmers who own relatively larger plots of land are concentrated in the central region (Attavanich et al. 2019). As a result, each region has attempted to overcome income constraints in different ways.

The northeastern farmers transitioned from extensive to intensive land use, marked by an increased reliance on agrochemicals to mitigate production risks, albeit with potentially adverse environmental consequences. In the central provinces, irrigation systems contributed to income stabilization until the 1970s. Nevertheless, both exertions failed to provide sufficient measures to break the vicious circle of impoverished agricultural households (Faysse et al. 2020). One recently emerging opportunity is facilitating technology adoption to alleviate productivity constraints, and yet, the high costs associated with policy implementation still need to be addressed.

A particular innovative framework that has had a widespread impact on rice production is the government-led promotion of the Sustainable Rice Platform (SRP), a global alliance convened by the United Nations Environment Program (UNEP) and the International Rice Research Institute (IRRI). The initial rationale of the policy was to secure production of the staple crop and follow global consensus to preserve natural resources; therefore, the standard of SRP is designed to increase resource efficiency through the value chain. The certification standard mainly requires farmers to use an adequate level of agrochemicals and enhance the economic viability of production through community engagement (SRP 2020). Rice that adheres to SRP standards can be marketed locally and globally, benefiting farmers with broader market opportunities and potentially higher profits.

Especially in the northeast region, initial programs of SRP version 1.0 provided authorized training sessions from 2013 to 2017 (Sae-heng et al. 2021). In our sample, 34% of households adopted SRP, which is significantly higher in the northeast region, making it reasonable to expect behavioral changes induced by knowledge transfer. On the contrary, it is notable that SRP farmers are generally inclined to absorb new information compared to conventional farmers, in that they should pay particular attention to policy arrangements and standards. Therefore, a causal relationship between additional information and SRP adoption must be clarified.

Figure 2 visualizes the outline of social networks from the sample. Each node is a household that provides or receives agricultural information within the network. Among 1,392 surveyed households, the analytical network comprises 1,012 nodes and 659 directed ties, capturing only those with at least one recorded knowledge exchange tie. While all households, including the 380 without any reported connections, were retained in the regression analysis as isolated nodes, the network structure analysis is based on the subgraph of connected individuals. The resulting network is sparse (density = 0.00064), with a diameter of 6, a global clustering coefficient of 0.151, and an average local transitivity of 0.298. Degree assortativity is low (0.038), indicating limited homophily in connectivity.

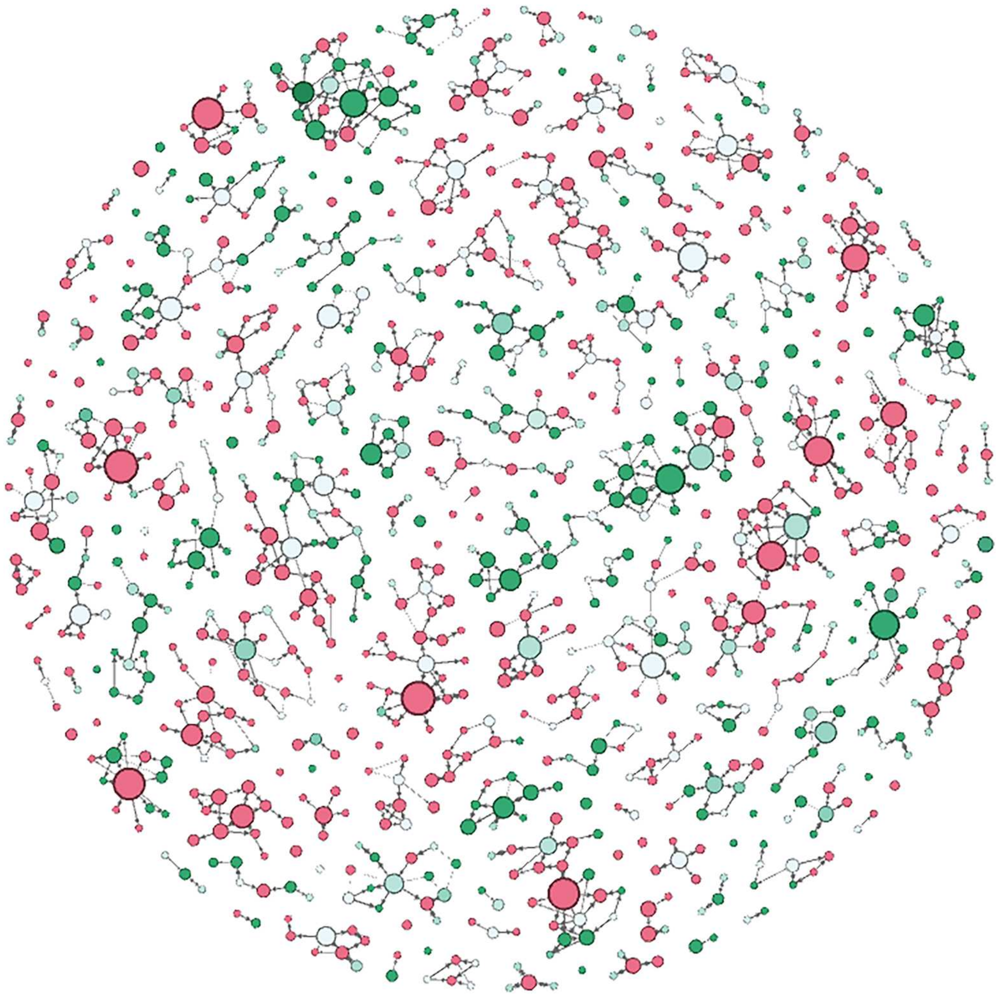

Figure 2. An example of degree centrality network.

Note: The color of the nodes represents the participation of the SRP standard in production (red nodes are conventional farmers while green nodes are SRP farmers), and the size of the nodes is proportional to one's weighted degree centrality. Edges represent information-sharing relationships.

Technologies

This study focuses on five technologies deemed crucial for widespread adoption in the region to promote agricultural development and stabilize farm income. These innovative approaches aimed at environmental sustainability are placed under the classification and definition of performance indicators presented in the SRP version 2.1, which were active during data collection (Figure 3).

Water resource management

Rice is a water-intensive crop, and continuous access to water resources decides whether cultivation in the dry season is possible. However, only farmers in the central areas benefit from irrigation, while smallholders in the northeastern lowlands still rely heavily on rainfed rice, suffering from low productivity (Suwanmontri, Kamoshita, and Fukai 2021). Alternatively, farmers can build a small-scale water management system, such as rainwater reservoirs, to adjust the level of water usage based on their expenses (Maneepitak et al. 2019). In our study, water management technology is defined as a more comprehensive concept encompassing the control of both total water productivity and quality. This includes Alternate Wetting and Drying techniques (AWD), rainwater harvesting, community-level irrigation systems, and water quality control (SRP 2020). In our sample, 29% of households adopted water management technologies, and 57% have access to an official irrigation system.

Straw baling and post-harvest management

Although reducing the carbon content has been a long-standing policy objective, most farmers follow the tradition of burning agricultural residuals on-site to minimize post-harvest efforts (Maneepitak et al. 2019). As Kanokkanjana and Garivait (2013) underline, the economic and environmental benefits attained from the alternative management of rice stubble opened a discussion to promote baling technology nationwide, while we find that 64% of the surveyed households adopt baling. However, using straw with additional labor or supplementary equipment poses a challenge for some farmers, particularly those who need more awareness of its long-term effectiveness. For instance, farmers could benefit from increased nutrient efficiency in farmlands, reducing livestock feeding costs, or generating extra revenue by selling straws to consumers. However, even farmers who recognize the advantages of baled straw often need help finding balers; thus, they fail to meet the social optimum because of market discord. The baling technique in our study is relaxed, including any efforts to gather rice stubble, whether using machinery or by hand, as opposed to the traditional slash-and-burn methods.

Crop rotation or cover crop (CRCC)

Production portfolio diversification is essential for avoiding production risk. Crop diversity and adaptive crop varieties can increase risk-adjusted returns and decrease GHG emissions. There is empirical evidence that 92% of all provinces that produced another major crop along with rice had a higher return than monoculture (Attavanich et al. 2019). However, for farmers who alternate two types of rice a year, the advantages of crop rotation may only sometimes lead to reduced risks or increased agricultural revenues. Otherwise, farmers can grow non-invasive cover crops to suppress weed growth and attain nitrogen fixation, associated with long-term production benefits. The dependent binary variable *CRCC* represents the adoption of rotational or cover crop practices, excluding the rotation within rice verities, as an indicator of innovative changes in the crop calendar. In our sample, 30% of the participants responded positively to this classification.

Fertilizer and pesticide control

Another driving force behind modern agricultural productivity is the invention of synthetic fertilizers and pesticides. However, two contradictory missions in developing countries still exist: encourage farmers to use agrochemicals to increase yield and lower the input amount to an adequate level for cost-effectiveness (Duflo, Kremer, and

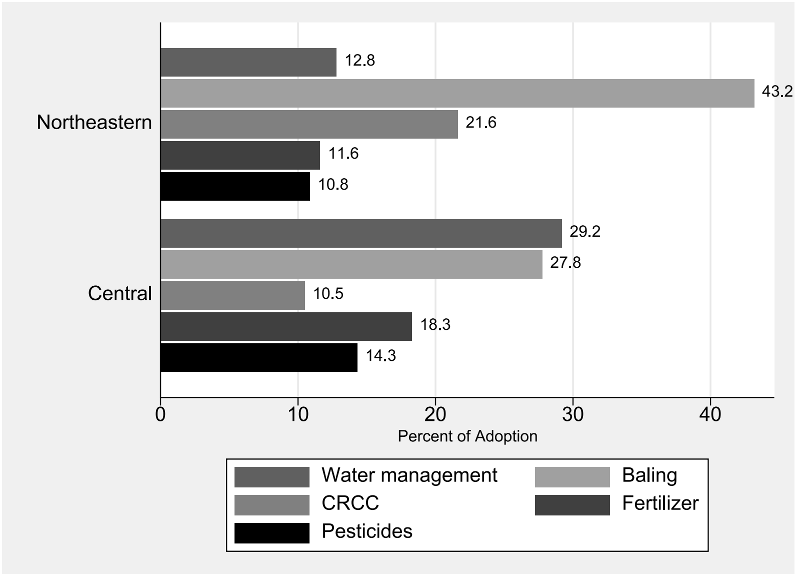

Figure 3. Percentage of technology adoption by regions.

Robinson 2011). In addition, Thailand has promoted fertilizer and pesticide control based on the chemical composition of their farmlands as a subordinate standard of SRP to reduce environmental risks. Thus, the introduction of agrochemical management not only restores the balance of Nitrogen, Potassium, and Phosphorus (NPK) ratios but also rectifies any inefficiency resulting from the irrational choice of input (SRP 2020). Our study examines this deliberate action aimed at reducing costs and increasing output, with the adoption rates for fertilizer and pesticide control at 23% and 20%, respectively.

Methods

Multivariate probit model

Each household reports whether it has adopted any of the five technologies above, and we assume that these binary decisions are correlated. Some technologies may reduce the additional investment requirements of others, which can be referred to as learning externality or path dependence (Foster and Rosenzweig 2010). Therefore, we use a multivariate probit model (MVP) to incorporate this complementarity, extending the binary choice model to several dependent variables that are defined simultaneously. Previous examples using the MVP have clarified how information from multiple sources can improve technology adoption (Hidayat, Noor, and Handono 2024; Jensen 2007; Mittal and Mehar 2016; Rahman and Chima 2015).

The general expression of MVP is written by the Greene (2012):

$$y_m^* = \mathbf{X}'_{\mathbf{m}}\boldsymbol{\beta}_{\mathbf{m}} + \epsilon_{\mathbf{m}}, \quad \mathbf{m} = \mathbf{1}, \ \cdots, \ \mathbf{M}, \quad [\epsilon_1, \epsilon_2, \ldots, \epsilon_{\mathbf{M}}] \sim \mathbf{N}_{\mathbf{M}}[\boldsymbol{0}, \ \boldsymbol{R}] \tag{1}$$

$$y_m = \begin{cases} 1 \ if \ y_m^* > 0 \\ 0 \ otherwise \end{cases}$$

where y_m^* is the latent variable for technology adoption, X_m is a vector of explanatory variables, β_m is a vector of parameters to be estimated, and ϵ_m is a random error term drawn from the M-variate normal distribution with an expression of variance-covariance matrix R in equation (2).

$$\boldsymbol{R} = \begin{bmatrix} 1 & \cdots & \rho_{1m} \\ \vdots & \ddots & \vdots \\ \rho_{m1} & \cdots & 1 \end{bmatrix} \tag{2}$$

The particular interest lies in the off-diagonal components of Σ, which indicates the unobserved correlation between the stochastic components of each decision (Young, Valdez, and Kohn 2009). For instance, $\hat{\rho}_{jm} > 0$ implies that the unobserved factor that increases the probability of adopting technology j is also positively affecting the probability of adopting technology m. Besides, the marginal probability of adoption on m^{th} technology and the joint probability of adopting all the technologies ($y_1 = \cdots = y_M = 1$) are expressed as below:

$$M\arg inal\ \Pr obability\colon\ \Pr(y_m = 1) = \boldsymbol{\Phi}(X'\beta_m)$$

$$Joint\ \Pr obability\colon\ \Pr(y_1 = \cdots = y_M = 1) = \boldsymbol{\Phi}_m(X_1'\beta_m,\ \cdots,\ X_m'\beta_m;\ \boldsymbol{R})$$

where $\Phi_m$denotes the M-variate standard normal cumulative distribution function. A practical approach to evaluating M-variate derivatives is constructed by Cappellari and Jenkins (2003) using the simulated maximum likelihood. For the analysis using STATA 18.0, the number of draws was set to 50 in *mvprobit*, above the square root of the sample size, to obtain consistent estimates.

Calculating the full spectrum of the marginal effect becomes extraordinarily challenging in MVP when 2^M cases of conditional probability for each outcome exist. However, Young, Valdez, and Kohn (2009) proposed an alternative method to simplify the calculation, specifically for the scenario starting from no outcome being selected ($y_1 = \cdots = y_M = 0$). In this case, the exponential of the index function, $(e^{\beta_{im}} - 1)*100$ can be used as a proxy to interpret the effect of x_i on the probability of m^{th} outcome relative to the probability of not adopting any technology, which is called the 'omitted category' (Rahman and Chima 2015).

Centrality measurements for social network

Notably, several of these technologies have significant learning curves, while their benefits extend beyond individual gains, with the potential to be shared across society. Social networks have emerged as a crucial factor in this context. These networks can facilitate knowledge exchange, foster collaboration, and most importantly, accelerate the adoption of these technologies (Abdul Mumin and Abdulai 2022; Bandiera and Rasul 2006; Weyori et al. 2018). By leveraging social connections, we can collectively drive positive change and enhance agricultural practices to achieve sustainable production. Therefore, this study uses network measurements to capture farmers' pre-existing status in the network, built-in as explanatory variables in our model.

In our network, each household is represented as a node, and a directed edge is drawn from one node to another when the respondent reports regular knowledge exchange. These edges reflect reported information-sharing relationships between farmers. To characterize the structural importance of each household within the network, we compute several centrality measures that capture the node's position in terms of connectivity and influence over information flow.

$$\textit{Degree Centrality}\text{: } \frac{d_i(g)}{n-1} \tag{3}$$

$$\textit{Betweenness Centrality}\text{: } Ce_i^B(g) = \sum_{k \neq j: i \notin [k,j]} \frac{P_i(kj)/P(kj)}{(n-1)(n-2)/2} \tag{4}$$

$$\textit{Closenes Centrality}\text{: } \frac{n-1}{\sum_{j \neq i} l(i,j)} \tag{5}$$

The most straightforward measure of position is *degree centrality*, which is the number of connections from i^{th} node (individual), $d_i(g)$ in equation (3), divided by $n-1$. This can be divided into two indicators, *indegree* and *outdegree*, based on the direction of connection. In addition, we use *betweenness centrality,* which indicates the importance of a node with respect to connecting to other nodes. In equation (4), $P_i(kj)$ is the shortest path between k and j that i lies on, and $P(kj)$ is the total number of paths between k and j. If the i node is on the most important (shortest) path between k and j, the ratio in the numerator converges to 1. Finally, *closeness centrality* indicates how easily a node can reach others. It is simply the inverse of the average distance between one node and any other node, where $l(i, j)$ is the number of nodes in the shortest path between i and j. This indicates the average shortest distance of one's overall connection to another, which can be interpreted as a larger scale or size of one's network when the closeness centrality is smaller. Figure 4 illustrates the network effect, where indegree and outdegree are the numbers of inward and outward information connections.

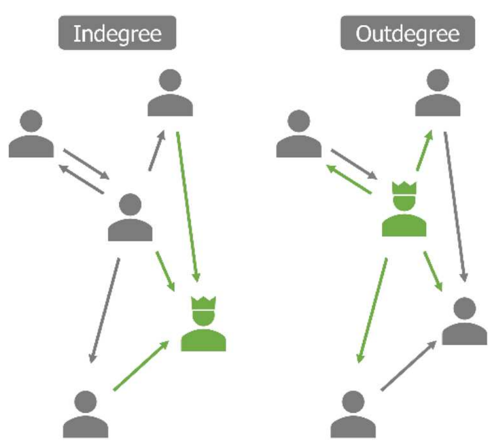

Figure 4. In-degree and out-degree networks.

Table 2. Average of network measurements by region.

Network Measurements	Full sample	Central	Northeast
Degree Centrality	2.07	1.98	2.1
Indegree	1.23	1.14	1.25
Outdegree	0.84	0.84	0.85
Betweenness	3.03	4.07	2.74
Closeness	0.19	0.18	0.19
N	1,392	311	1,081

On average, households have 1.23 connections for receiving agricultural information and 0.84 for giving information, for a total of 2.07 Degrees of Centrality, which does not differ across regions. The betweenness of households is significantly higher in the central region, meaning that central farmers are more central in connecting two or more heterogeneous groups. Simultaneously, there was rarely an average difference in closeness centrality (Table 2).

Procedure and data

Considering five SRP-themed technologies (m) and five centrality measures (j) of the network variable (NW), our estimation procedure for each household i can be written as.

$$\begin{aligned} Tech_{ijm} = {} & \beta_0 + \beta_1 \ln Land_i + \beta_2 Labor_i + \beta_3 SRP_i + \beta_4 \ln Ag\mathrm{Rev}_i + \beta_5 \ln Liv\mathrm{Rev}_i \quad (6) \\ & + \beta_6 \ln OffInc_i + \beta_7 \ln Debt_i + \beta_8 \ln Savings_i + \beta_9 \ln \Pr operty_i + \beta_{10} Irrigation_i \\ & + \beta_{11} Central_i + \beta_{12} Age_i + \beta_{13} Education_i + \beta_{14} Gender_i + \beta_{15} Usefu\ln ess_{im} \\ & + \sum_k \delta_k \Pr ovince_i + \alpha_j NW_{ij} + \epsilon_i, \quad j = 1, \ldots, 5, \; m = 1, \ldots, 5 \end{aligned}$$

where $Tech_{ijm}$ is the dependent dummy variable for technology adoption (m); categorized explanatory variables of production (*land*, *labor*, *SRP*), financial status (*AgRev*, *LivRev*, *OffInc*, *Debt*, *Savings*, and *Property*), accessibility to a resource (*Irrigation* and *Central* dummy), household characteristics (*Age*, *Education*, *Gender*, and *Usefulness*), and province dummy variables are included as factors affecting technology adoption. NW_{ij} is network measurement for each centrality (j), α, β, and δ are parameters to be estimated, and ϵ_i is assumed to have a multivariate normal distribution.

Agricultural production variables include total land size (ha) and number of effective labor (persons/household), which are the most significant input factors (Attavanich et al. 2019). Effective labor in this model represents the number of people in each household aged 15–65 years who can substantially contribute to on – and off-farm labor. Another element in production is the method of production, represented as a dummy variable for SRP, which is distinguished from conventional production ($SRP = 0$). This comprehensive strategy for long-term sustainability potentially implies comparable activeness in entering the broader market. In our sample, households have 3.63 ha of total average farmland, including rented land, while having 2.46 people for income-generating activities (Table 3).

Another crucial factor in innovative decisions is a household's financial status, and we divide the source of income into three categories. On-farm income, comprising agricultural and livestock revenue, reflects primary occupations, whereas supplementary off-

Table 3. Descriptive statistics.

Variables	Mean	SD	Min	Max
Production				
Land (ha)	3.63	(2.87)	0.24	33.6
Labor (person)	2.46	(1.26)	0	6
SRP (Yes = 1, No = 0)	0.34	(0.47)	0	1
Financial Status				
AgRev	88,249.87	(157,257.15)	0	2,328,350
LivRev	22,417.75	(74,031.49)	0	1,500,000
OffInc	44,169.55	(129,783.56)	0	4,000,000
Debt	263,489.58	(375,554.08)	0	4,030,000
Savings	37,538.37	(145,321.38)	0	4,300,000
Property	719,831.17	(809,049.20)	0	7,438,000
Accessibility				
Irrigation (Yes = 1, No = 0)	0.35	(0.48)	0	1
Central (Yes = 1, No = 0)	0.22	(0.42)	0	1
Household Characteristics				
Age (year)	59.44	(9.97)	24	97
Education (1-7)	1.77	(1.30)	1	7
Gender (Female = 1, Male = 0)	0.37	(0.48)	0	1
Expected Returns				
Usefulness of Water Management	4.95	(2.19)	1	7
Usefulness of Baling	5.65	(1.88)	1	7
Usefulness of CRCC	5.16	(2.13)	1	7
Usefulness of Fertilizer Control	5.06	(2.17)	1	7
Usefulness of Pesticides Control	4.79	(2.27)	1	7
N	1,392			

farm income outlines the ability to relax financial constraints and leverage farm investments (Reardon, Delgado, and Matlon 1992). As employment patterns in Thailand have become spatially diffused, off-farm activities and remittances have become considerable factors in household income (Rigg and Salamanca 2011). Additionally, the magnitude of debt and savings in both formal and informal banking systems, in addition to the value of real estate (*property*), is significant for determining collective wealth. Especially pervasive debt accumulation threatens agricultural households in Thailand, while access to external resources such as government-led initiatives can potentially alleviate its perilous effect (NSO 2019). The regional dummy 'Central' reflects this overall accessibility to modern infrastructure and resources.

The age of the household head generally represents the farmer's experience in agriculture and is related to their openness to new information or structural changes in behavior. In addition, the household head's education level is also pertinent to their risk attitudes, considering the cost of technology adoption (Mittal and Mehar 2016). The gender of the household head is included as a general household demographic variable to incorporate differences in the overall decision-making structure. The average age of household heads is approximately 60 years, and more than 60% of household heads completed elementary or lower levels of education. Meanwhile, the level of education is significantly higher in central villages, and 37% of households are female-headed, with no regional differences.

Moreover, Rahman and Chima (2015) found that personal traits, such as attitude or subjective appraisal toward change, can influence adoption behavior. Therefore, we incorporate the variable *Usefulness*, which estimates how beneficial technology is to farm production on a Likert scale (very little = 1, little = 2, somewhat less = 3, not much = 4, quite a lot = 5, a lot = 6, very much = 7). A positive attitude toward a technology would undoubtedly increase adoption, as it is closely related to prior information about one's farm management and the level of effort involved in adoption.

Network variables

Information is rarely costless or symmetrical (Aker 2010). In this context, social ties and the degree of centrality can be the determinants of adoption by reducing the learning cost of technology. The idea of network-based incentivization in information diffusion has substantial empirical evidence in Ghana (Abdul Mumin and Abdulai 2022; Conley and Christopher 2001), China (Cai, Janvry, and Sadoulet 2015), Vietnam (Hoang, Castella, and Novosad 2006), India (Beaman et al. 2021) and the US (Ramirez 2013). In Thailand's specific contextual framework, adopting various technologies related to improving household income requires knowledge of their long-term effectiveness, and disseminating information uniformly across all agricultural entities presents a challenge. Thus, it is predictable that the decision may depend on the relative position within the social network of farmers.

Results and discussion

Table 4 shows that adopting one technology positively correlates with the other decision-making process, except for some combinations, such as baling with water management and pesticide control. This provides the basis of the MVP approach to our correlated dependent variables.

The regression outcome of MVP without network measurements is presented in Table 5. The results of the likelihood ratio and Wald tests reject the null hypothesis that the correlation among the five technologies is zero at a significance level of 0.1%, indicating that the MVP model is more appropriate than the model that does not consider correlations among the five technologies.

Focusing on the production variables, the number of laborers significantly increases the adoption of the baling technique. In addition, participation in SRP standards positively influences the adoption of CRCC and pesticide management techniques. Thai farmers may choose to adopt a subset of the SRP standards. When the scores exceeded the threshold, an SRP label was obtained. In this regard, under autonomous decision-making, these two techniques are significantly related to achieving the final goal of SRP.

The total size of land as an input resource is entangled with innovative actions in that the introduction entails costs, with more extensive land holdings requiring more investment. Therefore, it may not be asserted that the scale of farmland positively influences

Table 4. Correlations between technology adoption behavior.

Technology	1	2	3	4	5
1. Water	1.00				
2. Baling	0.03	1.00			
3. CRCC	0.06*	0.20***	1.00		
4. Fertilizer	0.25***	0.10***	0.16***	1.00	
5. Pesticide	0.22***	0.01	0.12***	0.36***	1.00

** $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

Table 5. Determinants of technology adoption in multivariate probit without network measurements.

Parameter	Water Management		Baling		CRCC		Fertilizer Control		Pesticide Control	
Production										
ln Land	0.108	(0.068)	0.073	(0.065)	0.108	(0.064)	0.028	(0.069)	−0.036	(0.070)
Labor	0.036	(0.034)	0.107**	(0.035)	0.010	(0.033)	0.007	(0.034)	0.011	(0.040)
SRP	0.077	(0.086)	0.006	(0.089)	0.317***	(0.081)	0.157	(0.087)	1.343***	(0.098)
Financial Status										
ln AgRev	0.044***	(0.012)	0.034**	(0.011)	0.010	(0.011)	0.063***	(0.014)	0.049***	(0.013)
ln LivRev	0.000	(0.009)	0.067***	(0.009)	0.031***	(0.009)	0.007	(0.009)	0.001	(0.010)
ln OffInc	0.005	(0.008)	0.001	(0.009)	−0.007	(0.008)	0.009	(0.008)	−0.009	(0.009)
ln Debt	0.028**	(0.010)	0.013	(0.009)	0.018	(0.010)	0.007	(0.010)	−0.003	(0.010)
ln Savings	0.002	(0.008)	−0.012	(0.008)	0.012	(0.008)	−0.004	(0.008)	0.000	(0.009)
ln Property	−0.008	(0.019)	0.001	(0.017)	0.012	(0.019)	0.035	(0.025)	0.008	(0.022)
Accessibility										
Irrigation	0.284**	(0.101)	0.011	(0.104)	0.090	(0.098)	0.061	(0.103)	0.063	(0.109)
Central	0.057	(0.212)	0.898***	(0.244)	−0.733**	(0.260)	0.573*	(0.234)	0.130	(0.247)
Household Characteristics										
Age	−0.001	(0.004)	−0.003	(0.005)	0.012**	(0.004)	0.002	(0.004)	−0.003	(0.005)
Education	0.027	(0.030)	0.059	(0.034)	0.136***	(0.030)	0.093**	(0.030)	0.074*	(0.033)
Gender	−0.073	(0.090)	−0.136	(0.089)	−0.059	(0.084)	0.035	(0.088)	−0.050	(0.097)
Expected Returns										
Usefulness	0.296***	(0.027)	0.321***	(0.025)	0.284***	(0.025)	0.262***	(0.022)	0.314***	(0.023)
Province Dummy	yes									
N	1,392									

Note: *Standard error in parentheses, no constant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

*Likelihood ratio test of $\rho_{12} = \cdots = \rho_{54} = 0$: Log pseudolikelihood = −2842.14, Prob > Chi2 = 0.0000**** adoption despite the indication that large-scale farm operations are often correlated with exceptional performance and relatively higher revenue generation.

Regarding financial status, an increase in agricultural revenue is a significant factor in technology adoption, except for CRCC. This may come from a contradictory blind spot that the 'rotation of monoculture' activity is prevalent, where farmers practice double-cropping using a single variety. It needs further investigation; however, given that we control the total size of farmland, the difference in agricultural revenue is mainly led by the choice of a second crop. However, in our sample, farmers who produce twice a year do not diversify their commodities, implying that this may not correspond to crop rotation or cover cropping adoption. Additional income from livestock can significantly increase the adoption of baling and CRCC, which have synergetic benefits in utilizing crop residuals. On the contrary, the increase in debt can lead to adopting water management. At the same time, Donkoh, Azumah, and Awuni (2019) found that access to credit induces differing reactions to technology adoption. Thai farmers generally have access to credit; however, the amount they borrow varies, which can be interpreted as a liability that could be invested in facilities such as rainwater reservoirs.

When controlling for systematic differences between the central and northeastern regions, we find that accessibility to an irrigation system is separable from its actual usage. Farmers who use irrigation infrastructure are more likely to adopt water management technology. Attavanich et al. (2019) note that central regions have relative advantages in accessing water with more concentrated irrigation systems, while northeastern farmers rely on precipitation or use their water sources. Furthermore, farmers who practically use irrigation in the central region are inclined to manage water efficiency.

Regarding the household heads' demographic characteristics and attitudes, older and knowledgeable farmers tend to diversify their crops, and more educated farmers are also positively related to the adoption of fertilizer and pesticide control. Moreover, the self-reported usefulness of a technology increases its adoption.

However, building upon the understanding that farmers do not make decisions in isolation but rather learn from geographical and social neighbors (Conley and Udry 2010), it becomes evident that classic determinants of adoption, such as age, education, and access to credit, are valuable but fall short in explaining the influence of knowledge flows on technology adoption (Ramirez 2013). To address this limitation, we incorporated social network indices and found their significant impacts on technology adoption, as shown in Table 6 (full version with other independent variables is presented in Table 7, Appendix). The significance and magnitudes of the coefficients of the existing explanatory variables remain consistent; however, they present network-related factors that influence adoption in the context of different agricultural practices. The five columns show that the network factors have varying degrees of influence on the adoption of these practices.

A household with more channels connected to others (a higher degree of centrality) has a positive coefficient for adopting water management, crop rotation or cover crops, fertilizer, and pesticide control. It suggests the existence of a network effect that possibly increases the adoption of most technologies, which is in line with several theoretical and empirical literature (Abdul Mumin and Abdulai 2022; Cai, Janvry, and Sadoulet 2015; Conley and Udry 2010). The subdivision of degree centrality transforms the network into a directed graph, and the promotional significance of the network for adoption remains consistent only in the outward connection of information. This, in part,

Table 6. Network measurements as additional determinants of technology adoption.

Network Measurements		Degree Centrality		Indegree		Outdegree		Betweenness		Closeness	
		β	SE	β	SE	β	SE	β	SE	β	SE
Water Management		0.077***	(−0.018)	0.080*	(−0.038)	0.089***	(−0.026)	0.01	(−0.006)	0.250*	(−0.121)
	PE	8.004		8.329		9.308				28.401	
Baling		0.007	(−0.02)	0.037	(−0.038)	−0.004	(−0.027)	0.008	(−0.004)	−0.08	(−0.116)
	PE										
CRCC		0.044*	(−0.018)	0.02	(−0.035)	0.071**	(−0.025)	0.004	(−0.004)	0.131	(−0.115)
	PE	4.498				7.358					
Fertilizer Control		0.053**	(−0.019)	0.045	(−0.037)	0.072**	(−0.026)	0.004	(−0.004)	0.248*	(−0.124)
	PE	5.443				7.466				28.146	
Pesticide Control		0.049*	(−0.021)	0.038	(−0.04)	0.065*	(−0.028)	−0.005	(−0.005)	0.148	(−0.129)
	PE	5.022				6.716					
$\chi^2(5)$		24.72 ($p = 0.0002$)		5.87 ($p = 0.3187$)		20.10 ($p = 0.0012$)		9.54 ($p = 0.0895$)		8.3 ($p = 0.1407$)	
Province Dummy		yes									
N		1,392									

Note: *Estimates from other explanatory variables are not included. PE indicates the partial effects. Standard error in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$*

resonates with the findings of Banerjee et al. (2019), who defined 'gossip nominees' in a network as good at transmitting information. Consequently, the diffusion centrality (out-degree) as a possible proxy of the 'gossip' is expected to partially explain the effectiveness of increasing diffusion and take-up of technology, and the result is significant for all technologies that are influenced by the network effect. Our result highlights the importance of directionality in information flows: farmers who are active information providers, rather than mere recipients, are more likely to adopt new technologies.

Farmers with higher closeness centrality, defined as the inverse of the average shortest path to all other nodes, are more likely to adopt water management and fertilizer control technologies. Closeness centrality reflects how quickly and efficiently a farmer can access or disseminate information within the network, since a shorter average distance enables faster spread of knowledge and influence (Newman 2018). In our context, this suggests that farmers who are more centrally located in the network, and thus closer to others, are well-positioned to pioneer the adoption; however, the interpretation should be approached with caution as the index is sensitive to network size and structure. While closeness centrality showed some positive coefficients, the joint Wald test indicates that its overall effect on technology adoption is not statistically significant. Therefore, the interpretation of closeness centrality should be approached with caution, and its policy relevance appears limited in this context.

In terms of approximated partial effect, as supposed by Young, Valdez, and Kohn (2009), one more network connection is associated with an 8% ($\approx (e^{0.077} - 1) * 100$) increase in water management. In addition, the effects are similar even when partitioned into indegree and outdegree. However, in the case of CRCC and agrochemical controls, each increased by 4.5~5.4% from the undirected network, while marginal impacts appeared to be more prominent in the network of giving information.

A higher probability of adoption is consistently demonstrated as the out-degree increases. This questions a common assumption in policies related to technology dissemination, which primarily considers how many individuals receive information. Instead, the outcome underscores the importance of individuals actively sharing and diffusing information through networks. While receiving information through various channels is crucial for individuals, the emphasis shifts to having a sufficient understanding of the benefits of the technology to introduce and promote it to others effectively. This proactive approach can facilitate information dissemination and accelerate technology adoption across networks.

Overall, the results indicate that a household's position within a network can significantly influence the adoption of agricultural technologies. Upon revisiting the results in the group of other explanatory variables, farms with higher agricultural revenue, larger loans, and equipped with irrigation systems may favor water management adoption. Irrigation has the most significant coefficient of adoption. However, it is plausible that the information given to and received from other farmers may relax the constraints of physical accessibility and increase water-use efficiency.

Meanwhile, the likelihood of conducting bailing and post-harvest management increases with an additional unit of labor, but there needs to be more evidence of a network. What has been informed by qualitative interviews with local experts reveals instances where some farmers opt to lease or outsource the baling equipment due to the high cost of personally owning the equipment, and a mismatch between the baler and farmer markets may hinder the dissemination of technology.

Thailand's rice production exhibits two distinct features. The first is the attributes or nature of technologies that lead to long-term economic and environmental benefits under the Sustainable Rice Platform (SRP) theme. In general, it is analogous to a group of terms, such as climate-smart, climate-adaptive, or sustainable agricultural practices, which demand additional financial resources and support from the government to farmers (Sardar, Kiani, and Kuslu 2021). Our study contributes to understanding the crucial determinants of such technologies by focusing on the significance of individual information networks, which can provide a contextual understanding for designing an appropriate policy. Additionally, we focus on the correlated nature of these technologies; that is, farmers would decide whether to adopt them based on their relative advantages or synergetic effects.

However, the cross-sectional nature of our data limits our ability to analyze temporal dynamics in technology adoption sequences. As demonstrated in Conley and Udry (2010), observational network studies face inherent challenges in disentangling social influence from homophily effects. While our findings align with experimental evidence of Bandiera and Rasul (2006), we acknowledge the design limitation in capturing how early adopters accumulate network centrality over time, or how technology bundling creates path-dependent adoption patterns. Future longitudinal research should investigate the synergistic effects of combined technologies, particularly how water management adoption might precondition subsequent fertilizer/pesticide control decisions through network-mediated learning.

Conclusion

Innovative technologies are expected to play a crucial role in protecting farm income in Thailand (World Bank 2022). In addition to the independent household and farm characteristics commonly used in analyzing an agricultural household model, our findings indicate that the structure and direction of social networks are critical determinants of agricultural technology adoption, even when several technologies are correlated. First, five critical technologies in rice production were identified, and we clarified the different scenarios of social interactions influencing decisions, such as degree centrality, betweenness, and closeness. The multivariate probit model results show that adopting agricultural technologies has positive correlations, though certain combinations (e.g. bailing with water management and pesticide control) deviate from this pattern.

In summary, we observe a shift in the information-sharing perspective for technology adoption. This somewhat counterintuitive finding – that the 'giving' network is more intricately connected to technology than the 'receiving' direction – implies a policy recommendation: efforts should not only identify information recipients but also recognize the attributes of effective knowledge providers. Policies to accelerate innovation diffusion would benefit from supporting farmers who serve as information hubs within their communities. Enhancing outward information flows, alongside efforts to increase network density or proximity, is likely to offer a more effective strategy for scaling up technology adoption.

Empirical findings reveal that higher outward network connectivity positively influences the adoption of water management practices, crop rotation or cover cropping, fertilizer application, and pesticide efficiency. These outward-oriented social connections, akin to 'gossipers' in the literature (Banerjee et al. 2019), are shown to enhance the uptake of agricultural innovations significantly. In contrast, while closeness centrality – indicating proximity to others in the network – has theoretical appeal, its marginal effects were not statistically significant in our analysis, and thus its practical implications remain limited.

With regard to physical input variables, additional labor units significantly increase the likelihood of adopting baling practices, whereas the influence of farmland scale on adoption remains inconclusive. Income-related factors also matter, though the type and source of revenue contribute differently to adoption decisions.

Overall, this study provides robust empirical evidence on the importance of network centrality and knowledge-sharing dynamics in the adoption of agricultural innovation. The findings underscore the need to recognize and support influential knowledge providers when designing agricultural extension strategies. By fostering strong social connections and promoting active information exchange, policymakers and practitioners can significantly enhance the diffusion of innovation, ultimately improving productivity and resilience in rural communities. While this study offers valuable insights, the cross-sectional nature of the data limits our ability to examine changes over time or establish definitive causal relationships. Future research should consider longitudinal approaches to better understand how evolving network structures influence the trajectory of technology adoption and rural development outcomes.

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